

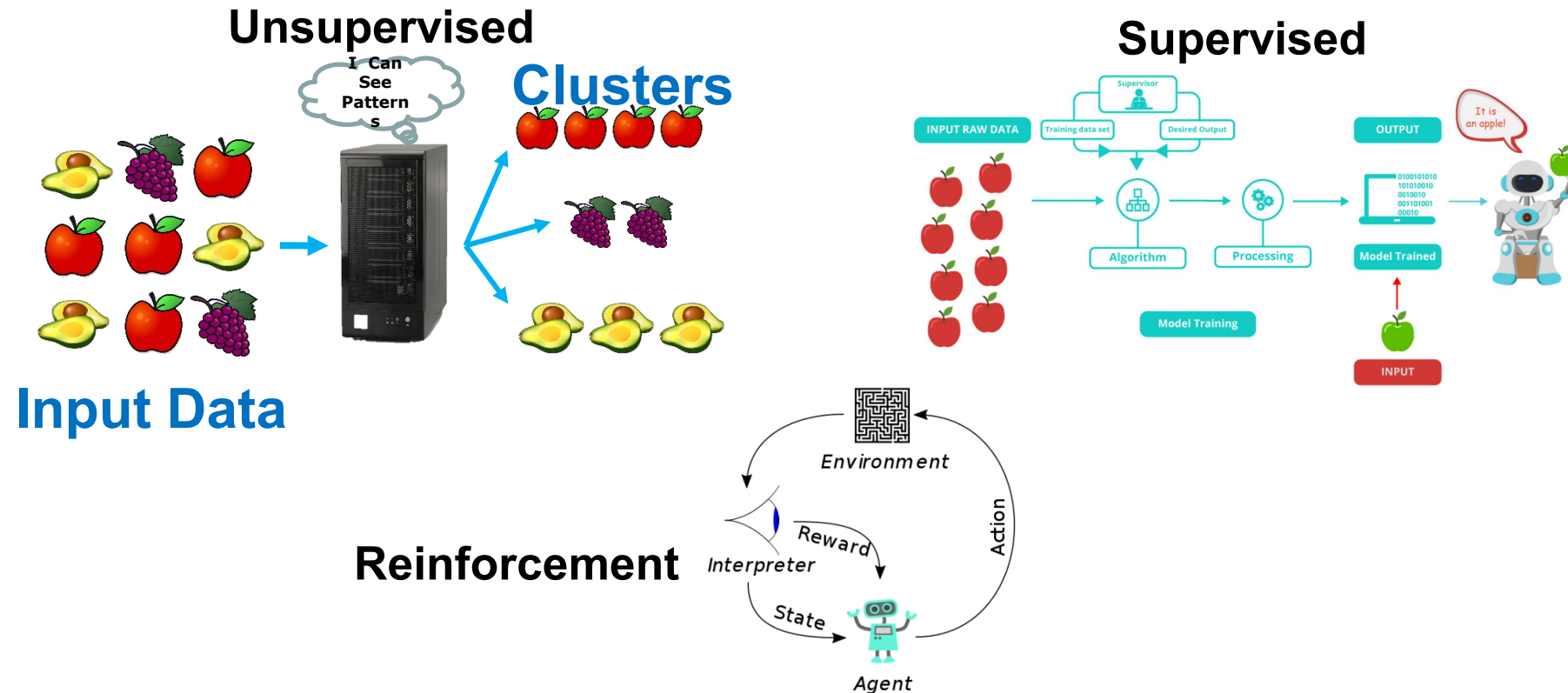


ECE 332 – Embedded Systems Lab

Lab 4: Intro to ML

Machine Learning

- Program learns from the data
- Types of machine learning: unsupervised, supervised, reinforcement



Unsupervised Learning

- Density estimation
 - Learning “what normally happens”
- Known input, unknown output
- Clustering: Grouping similar instances
- Example applications
 - Customer segmentation in CRM
 - Image compression: Color quantization
 - Bioinformatics: DNA mapping

Supervised Learning

- Train on **labeled** data
- Classify on future cases: Use rules to predict the output for future inputs
- Rules are learned during training

Supervised Learning: Training

- Needs a data set
- Example: CIFAR10
 - 10 Labels
 - 50,000 training images
 - Each image is 32x32 pixels
 - Each pixel has RGB values

airplane



automobile



bird



cat



deer



dog



frog



horse



ship

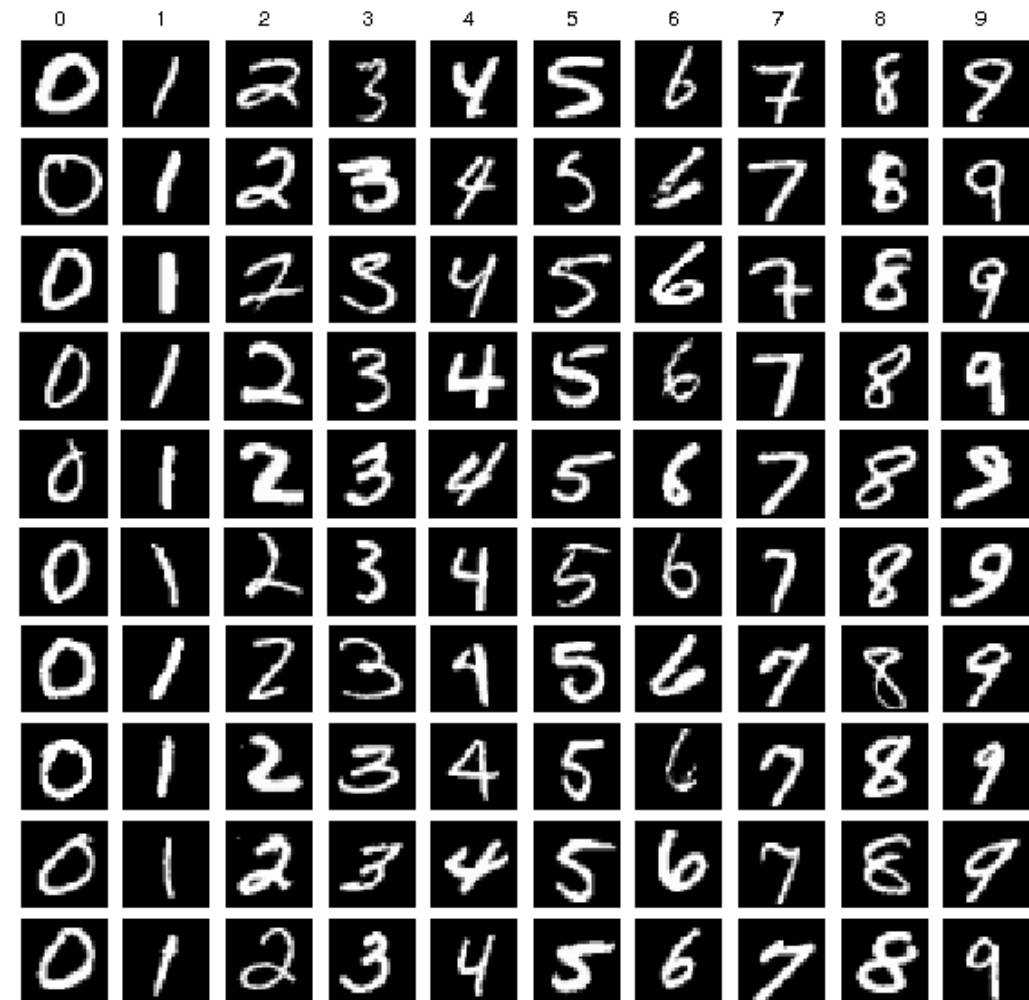


truck



MNIST data

- Recognition of numeric characters
- The lab is based on MNIST

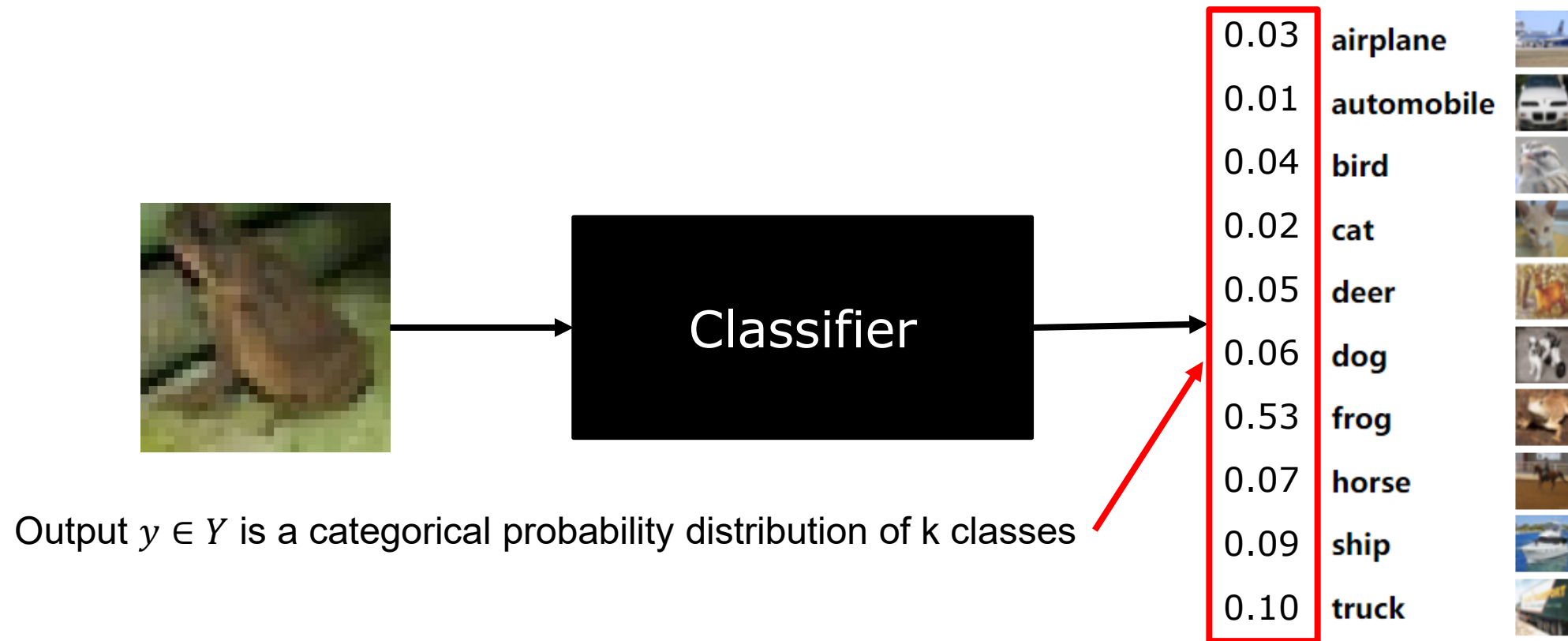


How does a program learn from data?

- Linear Classifier
- Neural Network

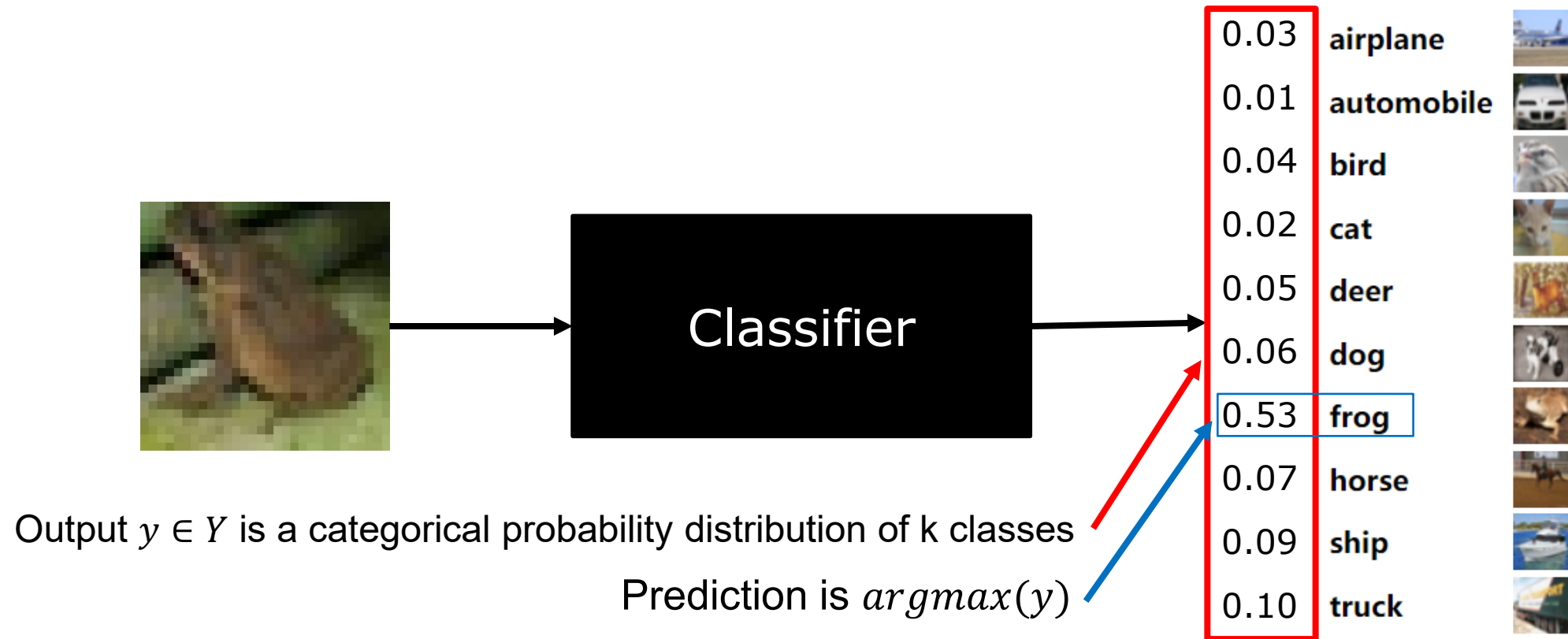
What Does a Classifier Do?

- Implements a function $X \rightarrow Y$, where $X \subseteq \mathbb{R}^d$, $Y \subseteq \mathbb{R}^k$



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Linear classifier



$$\boxed{f(x, W)} = \boxed{W} \boxed{x}$$

10×1 10×3072 3072×1

parameters, or “weights”

→ **10** numbers, indicating class scores

[32x32x3] array of numbers 0...1

Parametric approach: Linear classifier



$$\boxed{f(x, W)} = \boxed{W} \boxed{x}$$

10×1 10×3072 3072×1

$$\boxed{(+b)} \quad 10 \times 1$$

10 numbers, indicating
class scores

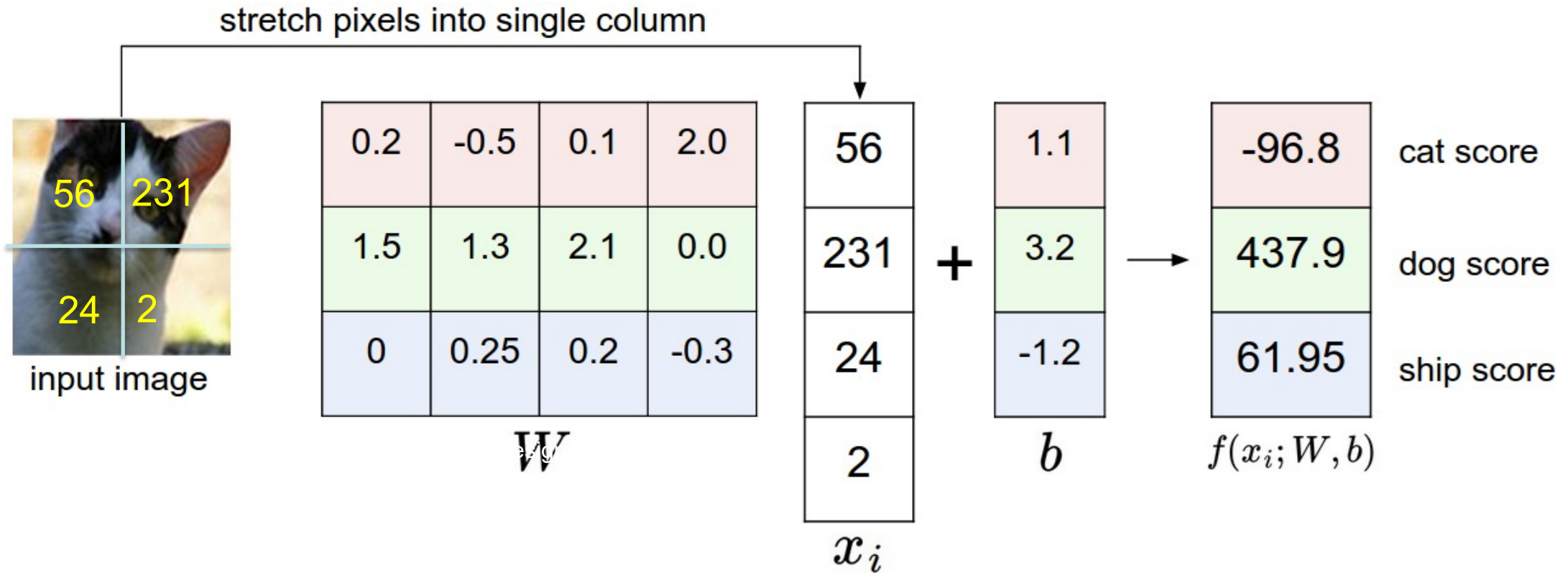
[32x32x3] array of numbers

parameters,
or “weights”

- **bias**, doesn't interact with the data; instead, just gives us some sort of data independent preferences for some classes over other.
- e.g., your dataset was unbalanced, and had many more cats than dogs, bias elements will adjust for that.

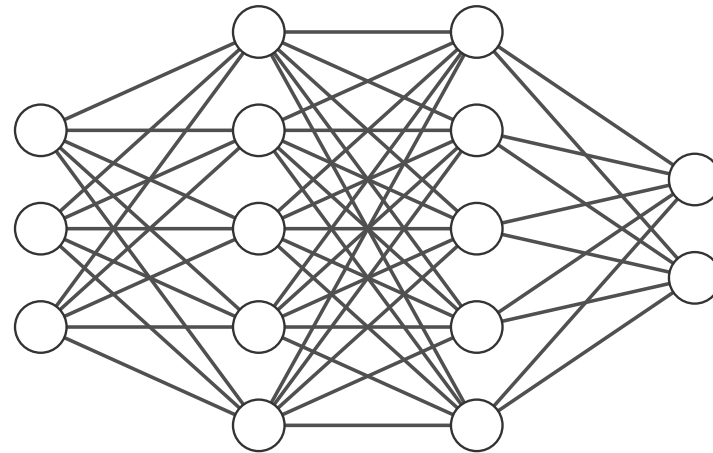
Example

An image with 4 pixels, and 3 classes (cat/dog/ship)

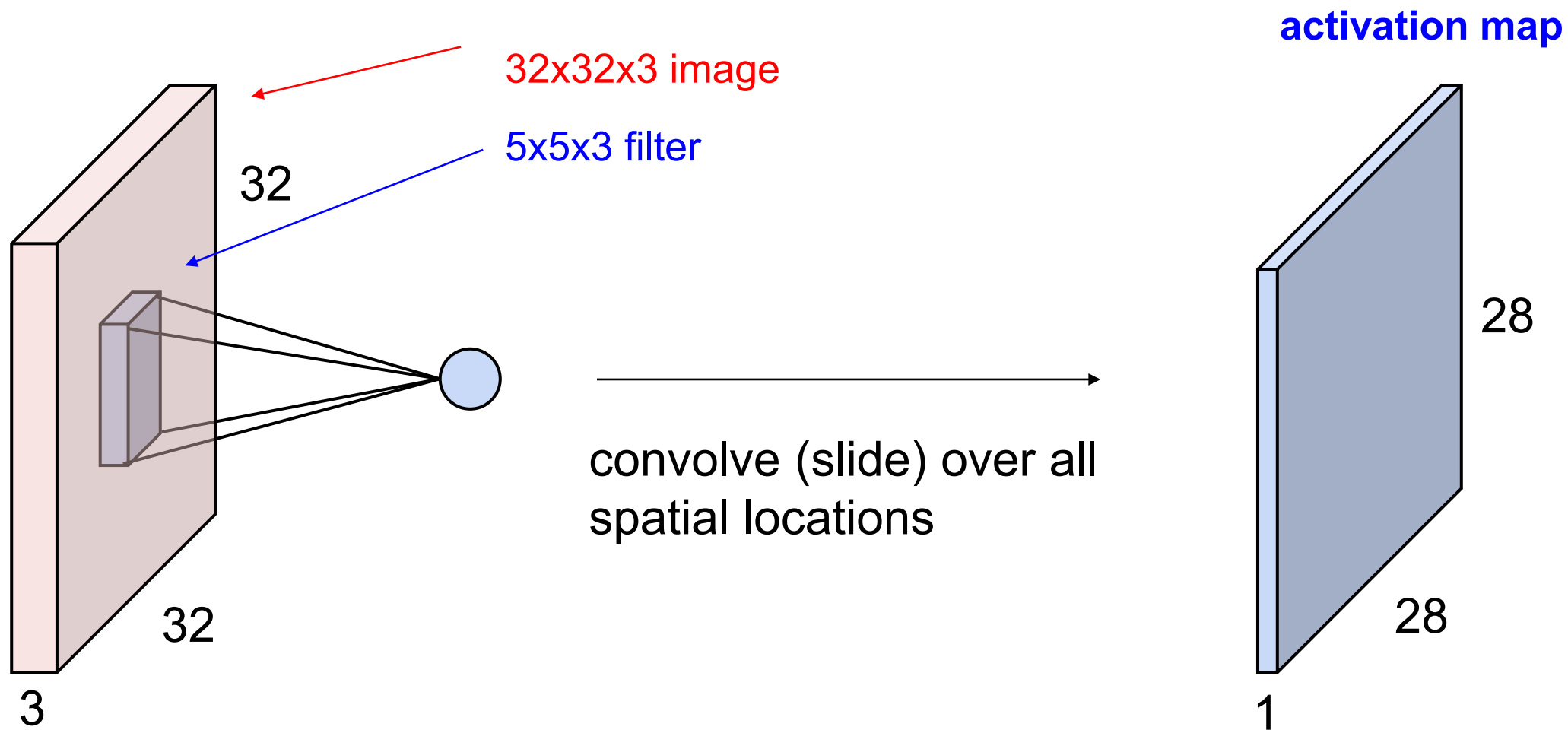


Convolutional Neural Network

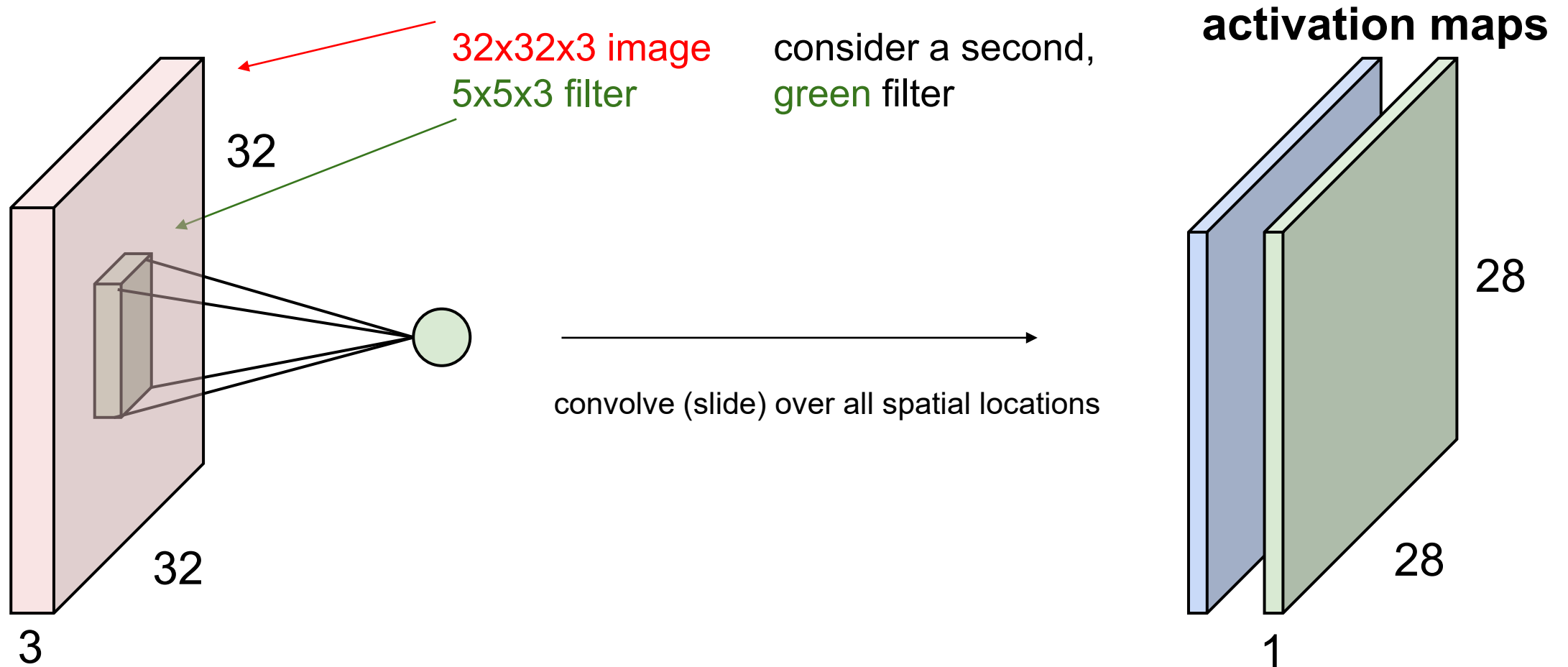
1. Convolution
2. Non-linearity
3. Add layers



Convolution Layer



Convolution Layer



Non-Linearity

- Instead of computing a linear combination

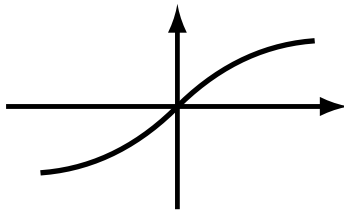
$$\text{score}(x, W)$$

- Add a non-linear function

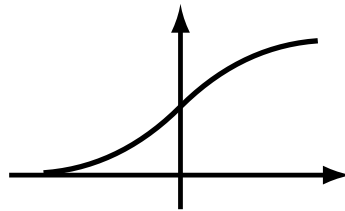
$$s_j = f(\text{score}(x, W))$$

- Popular choices:

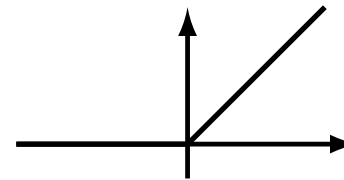
$$\tanh(x)$$



$$\text{sigmoid}(x) = \frac{1}{1+e^{-x}}$$

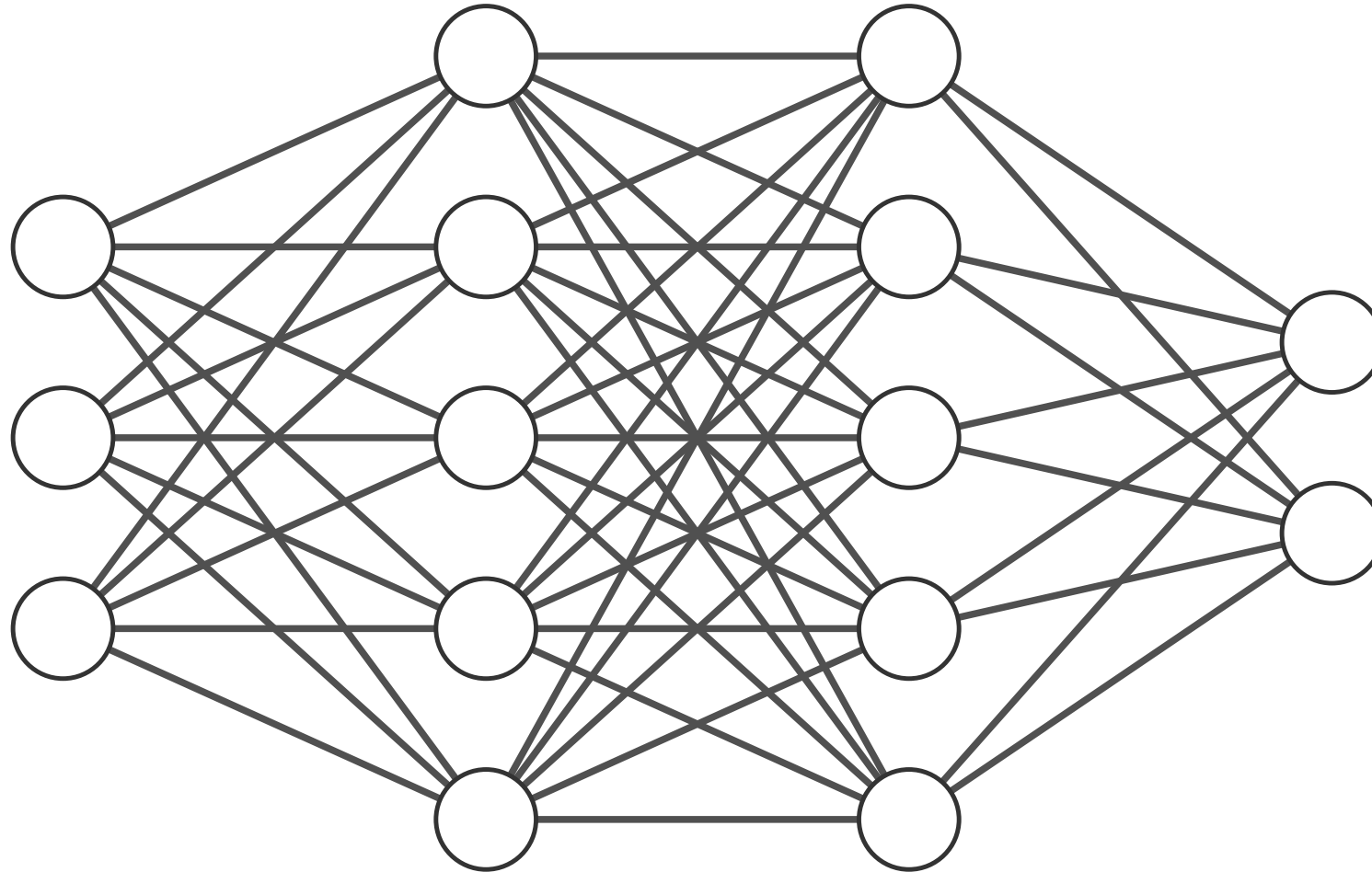


$$\text{relu}(x) = \max(0, x)$$



(sigmoid is also called the "logistic function")

Add layers



This lab

- Normally you train a network on data to learn the “filters”
- However, for this lab, you will be provided with a **pre-trained network**
- The pre-trained network involves **matrix operations** and **non-linear operations**
- Matrix operations will be performed in the FPGA
- Rest of the operations will happen in the CPU
- You will input an image of a handwritten digit, obtain the output “class” 0, 1, ..., 9

Exam Information

- On April 25, 2024, at ELAB 304 from 2:30PM-4:30PM
- No calculators, phones
- 2 by 2 seating

Exam format

- Embedded Processor
 - Short questions (2-3 points each)
- Memory and IO
 - Short questions
- Lab specific questions
 - Short questions
- Software Systems
 - Write a small program in C
- Operating Systems
 - Short questions
- Machine Learning
 - Short questions

Embedded Processor Architectures

- Definitions
- CPU architectures
- Hard and soft processors
- Non-functional requirements of embedded systems
- Altera SOPC, components and their uses
- IDEs used for labs
- NIOS II processor parameters (advantages of choosing one over the other)

Memory and I/O

- Various memory structures, what structures NIOS II supports
- MMU
- Interrupts and priority, SOPC interrupts
- interrupts
- I/O

Software Systems

- expression in C programming language
- data formats in C
- C programming question similar to the ones we coded in labs

Operating Systems

- Operating system
- Operating system - Scheduling
- Shared memory

Machine Learning

- Concepts
- Definitions